Opening Register Test: pre-registered spec

Written 2026-10-17, before any new dataset has been chosen or analysed. Nothing below may be changed after the first run on new data. Changes made later must be listed in a "Deviations" section.

Claim being tested

Novels open in a shared "opening register" that differs from the rest of the same book. In the original corpus (100 Victorian/Edwardian novels, 33 authors x 3 books, step 1 of the verification work) 13 of 24 features shifted in the first 1,500 words, and dialogue share was lower in the opening (about 17 points below words 6,000-12,000, lower in 30 of 33 authors).

This is a generalization test: the new dataset will NOT be late 19th / early 20th century. Both outcomes are informative. Modern novels often open in dialogue or action, so the dialogue prediction is the one most at risk. It stays as written.

Test (same as opening_register_scan.py)

- Trim front matter automatically; narrative start = first chapter heading followed by real prose.
- Opening = first 1,500 words of the narrative. Reference = the rest of the same book (300-word windows).
- Per book, z = how far the opening sits from the book's own windows, in window-to-window SDs.
- Author = unit of analysis (average z over the author's books). Sign-flip test across authors.
- Only the 13 pre-chosen features are tested, with BH-FDR across those 13 ($q < 0.05$, one-sided in the predicted direction).
- Dialogue is tested separately (not counted in the 13): words inside quotation marks per window, lower in the opening than in the rest of the book, same test and threshold.

Predicted directions (frozen)

- Higher in the opening (9): sentence length, word length (letters), "the", "of", sentence-length spread, syllables per word, "in", "and", "was".
- Lower in the opening (4): one-syllable words, "to", "but", type-token ratio (MATTR).
- Dialogue share: lower in the opening.

Verdict rule (frozen)

- **Confirmed:** at least 9 of 13 features pass in the predicted direction AND dialogue is lower in the opening.
- **Partial:** 5 to 8 features pass, or 9+ pass but dialogue does not.
- **Not found:** 4 or fewer features pass.

The other 11 original features are reported in a separate table labelled exploratory. They do not count toward the verdict.

Inclusion rules (frozen)

- Each book at least 20,000 words after trimming.
- Each author at least 2 books.

- Author = filename text before the first underscore, or a supplied authors.csv.
- Books whose narrative start could not be found with confidence ("uncertain" trims) are excluded and listed in the report. A book can be added back by listing its first sentence in starts.csv.
- Dialogue: quote style detected per book (double vs single, straight and curly). Books with no detectable dialogue are excluded from the dialogue test only. Detected style is listed per book.
- Every exclusion and its reason is listed in the report.

Known limitations

- Plain text loses italics, so internal thoughts or dialogue in italics are not counted as dialogue.
- Translated authors mix translator style into author style.
- Smaller corpora have less power than the original (33 authors); "not found" may mean low power rather than absence.
- Placebo check (same test at random mid-book positions) is run on the new data as calibration.

Out of scope (for now)

Attribution accuracy, voice-onset estimation, chapter-start control. These may follow if the register is confirmed.

Code

All new code goes in a new folder, Opening_Register_Test/, directly under Honors Thesis Code/. The original voice_pipeline scripts stay untouched. The script takes --corpus-dir and --out.

Dataset (to fill in before the first run)

Source, era, number of authors, number of books, date chosen: TBD.

Deviations

None yet.